

Answer **all** questions in the spaces provided.

1. There are different types of material in use today. Their uses are determined by their properties.

(a) (i) Complete the table below by selecting the word from the box to identify the type of material from its description.

Each word may be used once, more than once or not at all.

[3]

polymer	ceramic	composite	alloy	metal
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Description	Type of material
hard, brittle, high resistance	
hard, high melting point, good conductor	
mixture of two or more elements	

(ii) The properties of materials depend on their structure.
Describe the structure of polymers.

[2]

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- (b) Metals are very useful materials in a variety of industries. One use is for making wings of modern aircraft.

Some properties of the metals that could be used are given in the table below.

Metal	Density (kg/m ³)	Stiffness (GPa)	Tensile strength ($\times 10^7$ Pa)
titanium	4 500	110	55
aluminium	2 700	69	10
vanadium	5 700	138	63
steel	7 800	210	40

Use the information from the table above to answer the following questions.

- (i) State which metal could be used to make the wings as flexible as possible. [1]

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- (ii) I State which metal could be used to make the wings as light as possible. [1]

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- II Give **one** reason for your answer. [1]

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- (iii) I State which metal could be used so the wings can withstand large forces. [1]

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- II Give **one** reason for your answer. [1]

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